

Notification Service Database Design

1. Overview

Purpose

The Notification Service is responsible for processing domain events and delivering notifications to platform users and clients. The service consumes events from RabbitMQ and sends notifications asynchronously.

Responsibilities

- Process booking events
- Send booking confirmation emails
- Send booking cancellation emails
- Send booking reminder emails
- Store notification history
- Track delivery status
- Support future SMS notifications

Consumed Events

BOOKING_CREATED

BOOKING_CANCELLED

BOOKING_REMINDER

2. Database Technology

Property	Value
Database	PostgresSQL
Service	Notification service
Tables	1

Aggregate	Notification
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3. Aggregate Root

Notification

Represents a notification generated by a business event and delivered through a communication channel. Examples, Booking Confirmation, Booking Cancellation, Booking Reminder

4. Table Design

Table Name

notifications

5. Table Structure

Column	Type	Required	Description
notificationId	Long	Yes	Primary key
bookingId	String	No	Related booking
recipient	String	Yes	Email or phone number
notificationType	Enum	Yes	EventType
notificationChannel	Enum	Yes	Email or Sms
subject	String	No	Email subject
message	String	Yes	Notification body

status	Enum	Yes	Delivery status
sentAt	Date Time	No	Delivery timestamp
createdAt	Date Time	Yes	Creation timestamp

6. Validation Rules

Field	Validation
recipient	Required
Message	Required

7. Data Ownership Rules

Notification Service Owns

Notification

Notification Service References

bookingId from Booking Service. No foreign key exists.

Notification Service Does NOT Own

User, Business, Service Catalog, Availability and Booking.